

Mass Calculations and Galois Comparison

FFGFT Quantum Numbers, Lepton Masses and
Galois Fields $GF(3^n)$ Compared

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Mass Calculations and Galois Comparison

Abstract

This document places the FFGFT mass calculations of the three lepton generations alongside the corresponding Galois structures in $\text{GF}(3^n)$.

The main result: $(m_\mu/m_e)^2 = 43200$ is exact and identical in both frameworks — FFGFT from quantum numbers (r_i, p_i) , Galois from group orders $|\text{GF}(9)^*|^2 \cdot 5^2 \cdot |\text{GF}(27)|$. The equality is not a numerical coincidence but an algebraic identity from which $\xi = 4/30000$ follows as a Galois number **[K]**.

The remaining deviation of bare masses from experiment ($\sim 0.5\%$) is neither a measurement error nor K_{frac} ; it is an empirical discrepancy. The candidate is the fractal-recursive correction of higher order from the torus winding numbers; the physical attribution is open. Verification scripts: `pruef_330`, `pruef_331`.

.1 FFGFT Mass Formula and Quantum Numbers

The mass of fermion i in FFGFT (Doc. 006):

$$m_i^{\text{bare}} = r_i \cdot \xi^{p_i} \cdot v, \quad v = 246 \text{ GeV}, \quad \xi = \frac{4}{30000} \quad (1)$$

The quantum numbers (r_i, p_i) follow from the winding numbers $(n_{\theta,g}, n_{\phi,g})$ of the T^4 torus (Doc. 317):

$$r_g = \frac{n_{\phi,g}^2}{n_{\theta,g}}, \quad p_g = \frac{3}{2} \left(\frac{2}{3} \right)^{g-1} \quad (g = 1, 2, 3) \quad (2)$$

The values $3/2, 1, 2/3$ form the geometric sequence with $p_1 = D/2$ and ratio $2/3$ (Docs. 158, 018).

Table 1: FFGFT quantum numbers of leptons (Doc. 006, 317)

Lepton	Gen. g	n_{θ}	n_{ϕ}	r_i	p_i
Electron	1	3	2	4/3	3/2
Muon	2	5	4	16/5	1
Tau	3	9	5	25/9	2/3

.2 Lepton Masses: Bare Prediction vs. Experiment

Table 2: Lepton masses: FFGFT bare vs. PDG experiment

Lepton	m_i^{bare} (MeV)	m_i^{exp} (MeV)	Dev. (%)	Correction
Electron	0.5050	0.5110	−1.18	fractal-recursive
Muon	104.960	105.658	−0.66	fractal-recursive
Tau	1783.4	1776.9	+0.37	fractal-recursive

The bare prediction lies systematically 0.5–1.2% from experiment. This deviation is *not*:

- Measurement error (PDG errors $< 10^{-4}$ %)
- K_{frac} correction (K_{frac} cancels exactly in ratios, Doc. 070)

It is an empirical discrepancy: for the square $(m_{\mu}/m_e)^2$ the residual is $43200 - 42753.1 = 446.9$, far beyond any measurement uncertainty (Doc. 352). The candidate is the **fractal-recursive correction of higher order** from the torus winding numbers; the physical attribution is open.

K_{frac} is fixed by the comparison with the fine-structure constant (Doc. 070); the agreement of α itself is a property of the anchor E_0 :

$$\alpha = \xi \cdot E_0^2 / K_{\text{frac}}, \quad E_0 = \sqrt{m_e m_{\mu}} \approx 7.348 \text{ MeV} \quad \Rightarrow \quad \alpha^{-1} = 137.06 \text{ (dev. 0.02 \%)} \quad (3)$$

.3 Mass Ratios (K_{frac} -free)

In ratios, K_{frac} cancels exactly (Doc. 070):

$$\frac{m_{\mu}}{m_e} = \frac{r_{\mu}}{r_e} \cdot \xi^{p_{\mu}-p_e} = \frac{12}{5} \cdot \xi^{-1/2} = \frac{12}{5} \cdot \sqrt{7500} = 120\sqrt{3} \approx 207.846 \quad (\text{exp: } 206.768, \text{ dev. } 0.52 \%) \quad (4)$$

Table 3: Mass ratios: FFGFT bare vs. experiment

Ratio	FFGFT bare	Experiment (PDG)	Dev. (%)
m_μ/m_e	$120\sqrt{3} = 207.846$	206.768	+0.52
m_τ/m_e	3531.6	3477.2	+1.56
m_τ/m_μ	16.99	16.82	+1.04

.4 Galois Comparison

Core identity: $(m_\mu/m_e)^2 = 43200$

Squaring the mass ratio removes the irrational $\sqrt{3}$ (undefined in $\text{GF}(3^n)$ since $3 = 0$ there) and yields an exact integer identity **[K]**:

$$\underbrace{\frac{(r_\mu/r_e)^2}{\xi}}_{\text{FFGFT}} = \underbrace{|\text{GF}(9)^*|^2 \cdot 5^2 \cdot |\text{GF}(27)|}_{\text{Galois}} = 8^2 \cdot 25 \cdot 27 = 43200 \quad (5)$$

Table 4: FFGFT vs. Galois for $(m_\mu/m_e)^2$

	FFGFT	Galois
Input	$r_e = 4/3, r_\mu = 16/5, \xi$	$ \text{GF}(9)^* = 8, \text{GF}(27) = 27, 5$
Formula	$(r_\mu/r_e)^2/\xi$	$8^2 \cdot 5^2 \cdot 27$
Result	43200	43200
Status	[K]	[K]

The factors on the Galois side:

- $8 = |\text{GF}(9)^*| = 3^2 - 1$: multiplicative group of the 2nd generation
- $27 = |\text{GF}(27)| = 3^3$: field of the 3rd generation (all elements)
- $5 = \text{new prime}(|\text{GF}(81)^*| = 80)$: only new prime in $|\text{GF}(3^4)^*|$

Derivation of ξ from the identity

Since both sides equal the same number, ξ follows algebraically **[K]**:

$$\xi = \frac{(r_\mu/r_e)^2}{43200} = \frac{(12/5)^2}{43200} = \frac{144/25}{43200} = \frac{1}{7500} = \frac{4}{30000} \quad (6)$$

ξ is not a free constant — it follows necessarily from the Galois group orders and the winding numbers of the T^4 torus.

Winding numbers from $\text{GF}(3^n)$

Table 5: Winding numbers: derivation from $\text{GF}(3^n)$ [K]

Quantity	Value	Galois origin	Note
$n_{\theta,1}$	3	$\text{char}(\text{GF}(3^n))$	axiomatic
$n_{\phi,1}$	2	$ \text{GF}(3)^* $	mult. group of prime field
$n_{\theta,2}$	5	recursion $3 + 2$	also new prime factor of $ \text{GF}(81)^* = 80$
$n_{\phi,2}$	4	$\varphi(5)$	Euler- φ of prime factor
$n_{\theta,3}$	9	recursion $5 + 4$	also char^2
$n_{\phi,3}$	5	new prime factor of $ \text{GF}(81)^* $	

Recursion law: $n_{\theta,g} = n_{\theta,g-1} + n_{\phi,g-1}$. $\text{GF}(3^4)^* = 80 = 2^4 \cdot 5$ supplies the new prime factor 5, absent from $\text{GF}(3^1)^*$, $\text{GF}(3^2)^*$, $\text{GF}(3^3)^*$. 5 is new only relative to the prime factors of these levels: $|\text{GF}(27)^*| = 26 = 2 \cdot 13$ already brings the new prime factor 13, and the smallest prime factor of 80 is 2.

Complete comparison at ratio level

Table 6: Complete comparison at the ratio level

Method	$(m_\mu/m_e)^2$	m_μ/m_e	Dev. exp.
Experiment (PDG)	42 753.1	206.768	—
FFGFT bare $(r_\mu/r_e)^2/\xi$ [K]	43 200	207.846	+0.52 %
Galois $8^2 \cdot 25 \cdot 27$ [K]	43 200	207.846	+0.52 %

FFGFT and Galois give exactly the same bare value, so the deviation from experiment is the same in both frameworks. The comparison is made in the integer form $(m_\mu/m_e)^2$; the 0.52 % residual in m_μ/m_e is exactly half the quadratic residual and not an independent finding. The physical attribution of the residual is open (Doc. 352).

.5 Fine-Structure Constant from Galois Numbers

Two key identities

Combining two Galois identities eliminates ξ from the α -formula entirely.

Identity 1 (pruef_331, Doc. 317):

$$\xi = \frac{(r_\mu/r_e)^2}{|\text{GF}(9)^*|^2 \cdot 5^2 \cdot |\text{GF}(27)|} = \frac{144/25}{43200} = \frac{1}{7500} \quad (7)$$

Identity 2 (new observation):

$$m_e \cdot m_\mu = 53.991 \text{ MeV}^2 \approx 54 = |\text{GF}(3)^*| \cdot |\text{GF}(27)| \text{ MeV}^2 \quad (\text{dev. } 0.016 \%) \quad (8)$$

This holds for the experimental (fractally corrected) masses; bare masses give $0.505 \times 104.96 = 53.00 \text{ MeV}^2$ (dev. 1.85%).

Derivation — $|\text{GF}(27)|$ cancels

Substituting both identities into $\alpha = \xi \cdot E_0^2 / K_{\text{frac}}$:

$$\begin{aligned} \alpha &= \frac{1}{7500} \cdot 54 \text{ MeV}^2 / K_{\text{frac}} \\ &= \frac{|\text{GF}(3)^*| \cdot |\text{GF}(27)|}{|\text{GF}(9)^*|^2 \cdot 5^2 \cdot |\text{GF}(27)|} \cdot \frac{12^2}{5^2} / K_{\text{frac}} \\ &= \frac{|\text{GF}(3)^*| \cdot 12^2}{5^2 \cdot |\text{GF}(9)^*|^2 \cdot 5^2} \cdot \frac{1}{K_{\text{frac}}} = \frac{2 \cdot 144}{40000} \cdot \frac{75}{74} = \frac{27}{3700} \end{aligned} \quad (9)$$

$|\text{GF}(27)| = 27$ cancels between numerator and denominator.

Result

$$\frac{1}{\alpha} = \frac{3700}{27} = 137.037037 \dots \quad (\text{exp: } 137.035999, \text{ dev. } 0.00076 \% = 7.6 \text{ ppm}) \quad (10)$$

All inputs are Galois group orders:

- $2 = |\text{GF}(3)^*|$, $8 = |\text{GF}(9)^*|$, $27 = |\text{GF}(27)|$, $5 = \text{new prime}(|\text{GF}(81)^*|)$
- $12 = n_{\phi, \mu} \cdot n_{\theta, e} = 4 \cdot 3$ (winding numbers, Galois)
- $K_{\text{frac}} = 74/75$, $74 = 2 \cdot 37$, $37 \mid |\text{GF}(3^{18})^*|$ (Galois)

No ξ , no v_i nor m_e as explicit input. The only SI bridge is the MeV unit (implicit in $E_0^2 = 54 \text{ MeV}^2$). The agreement with the measured α is therefore a property of the anchor $E_0^2 = 54 \text{ MeV}^2$, not an independent prediction.

Table 7: Precision comparison: Galois results vs. experiment

Quantity	Galois result	Dev. from exp.
$(m_\mu/m_e)^2 = 43200$	43200 (exact)	+1.04% (bare)
$m_\mu/m_e = 120\sqrt{3}$	207.846	+0.52% (bare)
$1/\alpha = 3700/27$	137.037	0.00076% (7.6 ppm)

The α -formula is $\sim 700\times$ more precise than the bare mass ratio because the identity $m_e m_\mu = 54 \text{ MeV}^2$ holds for the experimentally measured (fractally corrected) masses.

The identity $m_e \cdot m_\mu \approx 54 \text{ MeV}^2$ is not an empirical coincidence. Doc. 011 derives $E_0 = \sqrt{m_e m_\mu}$ as the logarithmic average on the T0 torus geometry (there as the fundamental definition of E_0); fractally corrected it is $E_0 = \sqrt{m_e m_\mu / K_{\text{frac}}} = 7.398 \text{ MeV}$ (R72). The identification with $|\text{GF}(3)^*| \cdot |\text{GF}(27)|$ is a new Galois-side expression of the same quantity **[K]**. Verification script: `pruef_332_alpha_galois.py`.

Verification scripts

- `pruef_330_galois_massenverhaeltnis.py`: $(m_\mu/m_e)^2 = 43200$ (FFGFT = Galois), K_{frac} cross-check
- `pruef_331_xi_aus_galois.py`: $\xi = 4/30000$ fully derived from $\text{GF}(3^n)$

Note on AI assistance

This document was prepared with support from Claude (Anthropic, A-series). All algebraic calculations were verified by Python scripts with exact rational arithmetic (sympy). The physical interpretations and conclusions are by Johann Pascher (ORCID 0009-0000-6518-4064).

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